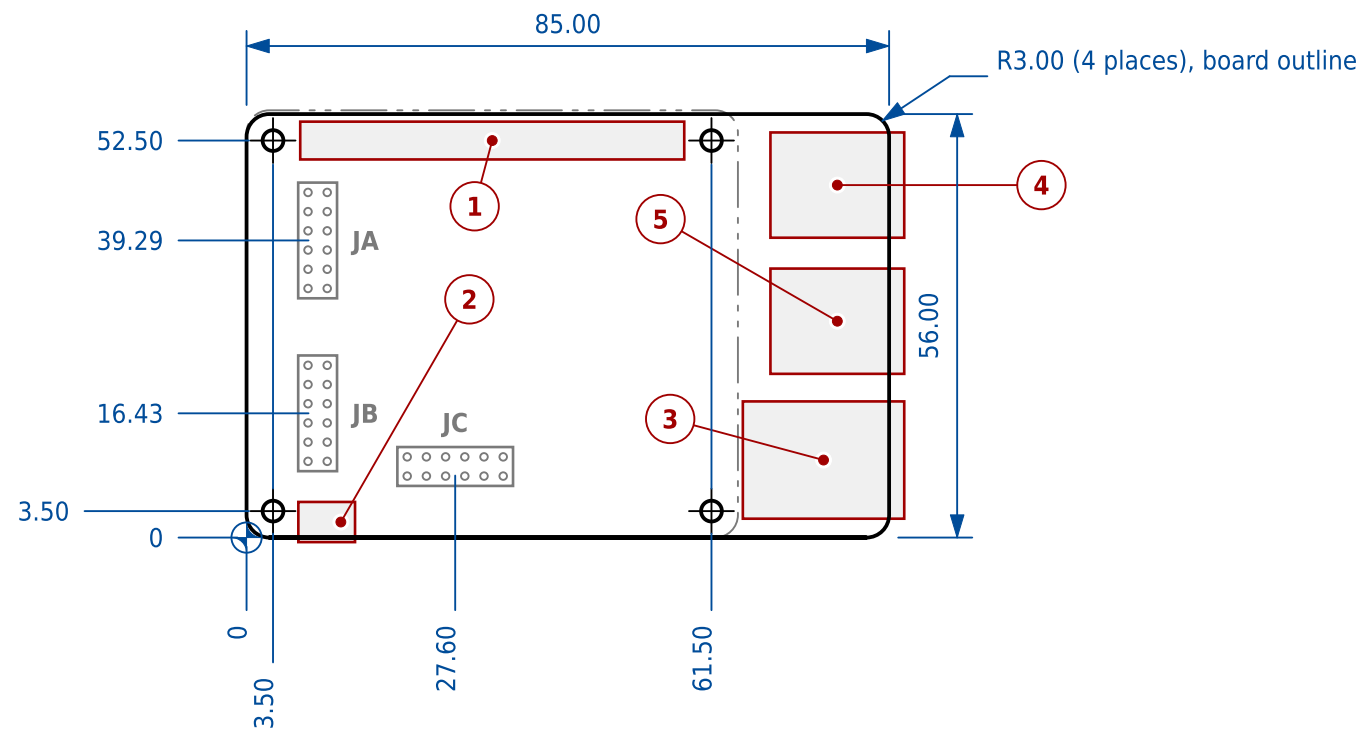




NOTES

1. All dimensions in millimetres. The datum symbol marks the origin: the board's lower-left corner, X right, Y up, seen from the component side.
2. Phantom outline is the Digilent Pmod HAT Adapter on the 40-pin GPIO header, drawn in this board's frame: its mounting holes coincide with this board's.
3. Connector outlines are the component body as drawn by Raspberry Pi Ltd, overhang past the board edge included.
4. Hole IDs are assigned by this drawing, bottom row first then left to right, and mean the same thing on every Raspberry Pi sheet here.
5. This sheet covers the Raspberry Pi 3 Model B and the Raspberry Pi 3 Model B+. Every connector position was read from both drawings separately and required to match, so the sheet covers both only while they agree.
6. The 40-pin GPIO header is fixed by the Raspberry Pi HAT specification and carries the same position on every Raspberry Pi sheet here. The drawings were read separately, agreed to 0.060 mm, and were snapped to one value.
7. Design mounting hole keep-outs to 6.2 mm diameter on every model: that is what the Raspberry Pi HAT specification (github.com/raspberrypi/hats) requires, and the largest figure the models themselves publish. The KEEPOUT column gives what this model's drawing shows, which is not always anything.
8. The hole diameter tolerance in the schedule, +/-0.05 mm, is quoted from this model's own drawing and governs in place of the hole diameter figure in the general tolerance block.
9. Hole positions are the 3.5 mm inset and 58 x 49 mm rectangle dimensioned on this model's drawing.

NOTES (continued)

10. Pmod host JC on the Pmod HAT Adapter overhangs the lower board edge, where these boards carry their HDMI and power connectors. Digilent's manual warns to fit the two standoffs opposite the 40-pin connector so JC's pins cannot touch the HDMI shell. The HDMI connectors are not drawn here.
11. Those host positions are DERIVED, not published: Digilent issue no mechanical drawing for the adapter. Good to about +/-0.75 mm. JA and JB face out of the left edge, JC out of the lower edge. The Pmod HAT Adapter sheet has the derivation.
12. Pmod host coordinates are given to three decimals, unlike the rest of this sheet. Rounded to two, adjacent hosts print 22.85 apart and contradict the 22.86 pitch dimensioned here.
13. Port names differ by family: JA, JB, JC on the adapter and Raspberry Pi sheets, by signal direction on the demo boards, PMOD 1 to 3 on the plate sheets. TT-MP-02 maps them.
14. Assembled envelope, connector overhang included, is 87.00 x 56.60 mm against a 85.00 x 56.00 mm board outline. The FEATURE SCHEDULE gives the extents.
15. Feature numbers are fixed across this family, so a number means the same part on every sheet. A part this board does not carry still has a row, marked as such.
16. GENERAL TOLERANCE in the title block is a normal board-house figure, not one any source here states. A tolerance quoted in the hole schedule is from the source drawing and governs.
17. Generated from the listed sources and not checked by a second party: there is no CHECKED field because nobody has signed it.

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	3.50	3.50	2.75 +/-0.05	not given
MT2	61.50	3.50	2.75 +/-0.05	not given
MT3	3.50	52.50	2.75 +/-0.05	not given
MT4	61.50	52.50	2.75 +/-0.05	not given

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	40-pin GPIO header	7.10 to 57.90	50.00 to 55.00
2	micro-USB power input	6.85 to 14.35	-0.60 to 4.71
3	Ethernet RJ45	65.65 to 87.00	2.50 to 18.00
4	USB 2.0 type A (upper pair)	69.30 to 87.00	39.65 to 53.57
5	USB 2.0 type A (lower pair)	69.30 to 87.00	21.65 to 35.57

PMOD HAT ADAPTER HOSTS

PORT	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
JA	left edge	9.400	39.290	8.130	45.640
JB	left edge	9.400	16.430	8.130	22.780
JC	bottom edge	27.600	9.400	21.250	8.130

LEGEND

- Board outline
- Component body, scheduled
- Adjacent part
- Dimension, extension and leader

SOURCES

1. Mechanical drawing: <https://datasheets.raspberrypi.com/rpi3/raspberry-pi-3-b-mechanical-drawing.dxf> - Raspberry Pi Ltd
2. Mechanical drawing: <https://datasheets.raspberrypi.com/rpi3/raspberry-pi-3-b-plus-mechanical-drawing.dxf> - Raspberry Pi Ltd, Raspberry Pi 3 Model B+
3. Mounting holes: <https://datasheets.raspberrypi.com/rpi3/raspberry-pi-3-b-mechanical-drawing.pdf> - Drawing note on the Model B drawing, quoted: 4x M2.5 MOUNTING HOLES DRILLED TO 2.75 +/- 0.05mm. The B+ drawing gives no hole size, but its geometry is identical.

RASPBERRY PI 3 MODEL B AND B+

85 x 56 mm, both models

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-85-g50a27ec	mm	1:1
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	RPI-01	A
MATERIAL		GENERAL TOLERANCE	
PCB, thickness not stated		edge +/-0.20 hole pos +/-0.10 hole dia per schedule	